

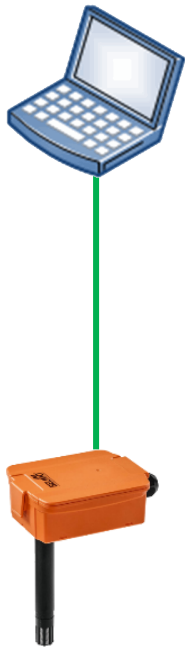
B2 - BACnet / Modbus (Advanced)

Modbus - Workshop I

Duct Sensor

B2 - Modbus - Workshop I – Sensor - Network

Modbus Poll



22DTH-15M

— Serial network (RS485)

B2 - Modbus - Workshop I - Sensor

1. Power your device.

 *See wiring diagrams in the datasheet*

2. Connect your device via Modbus RTU with your laptop.

 *Use RS485-USB adapter*

 *See wiring diagram in the datasheet*

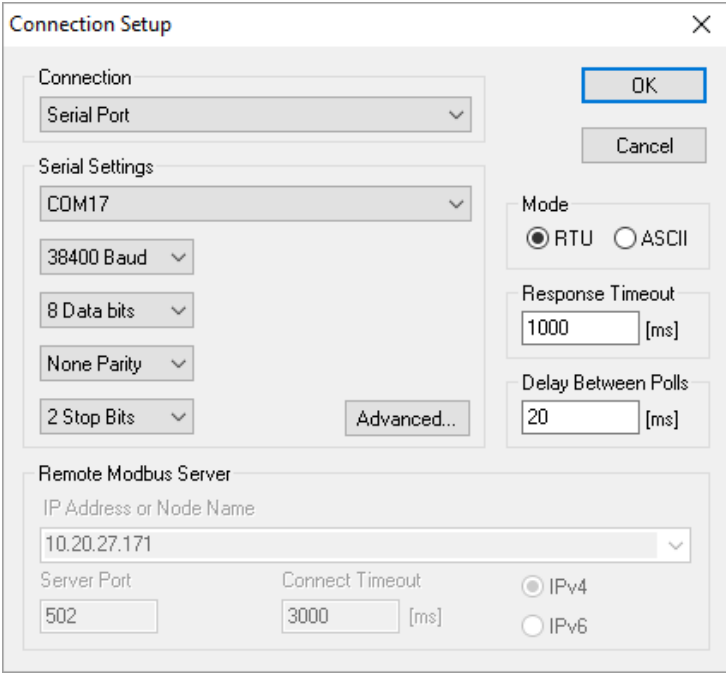
3. Configure your actuator according to network overview.

 *Find your appropriate configuration tool in the documents (LINK.10, Belimo Assistant 2, etc.)*

B2 - Modbus - Workshop I - Sensor

4. Start Modbus Poll. Open a **Serial Port Connection**.

- 💡 Check on which COM port your RS485-USB adapter is
- 💡 Pay attention to the Modbus settings on your actuator you just made



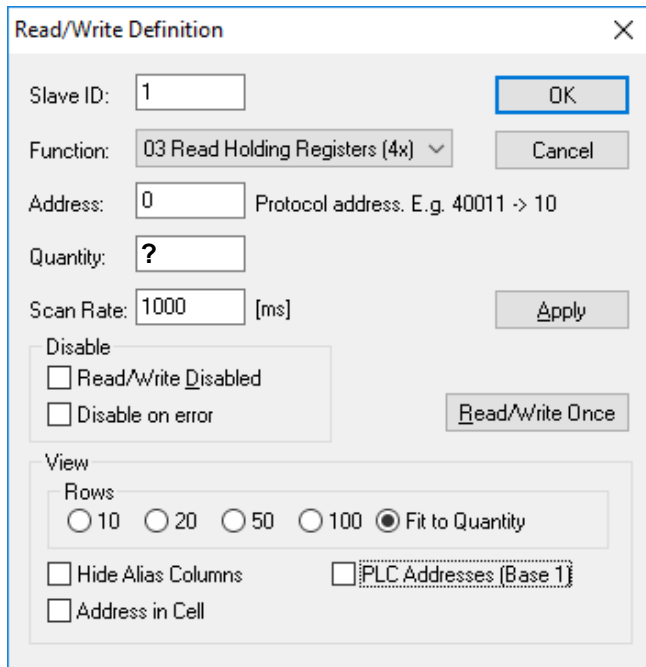
The image shows a 'Connection Setup' dialog box with the following settings:

- Connection:** Serial Port
- Serial Settings:**
 - COM17
 - 38400 Baud
 - 8 Data bits
 - None Parity
 - 2 Stop Bits
- Mode:** RTU (selected), ASCII
- Response Timeout:** 1000 [ms]
- Delay Between Polls:** 20 [ms]
- Remote Modbus Server:**
 - IP Address or Node Name: 10.20.27.171
 - Server Port: 502
 - Connect Timeout: 3000 [ms]
 - IPv4 (selected), IPv6

Buttons: OK, Cancel, Advanced...

B2 - Modbus - Workshop I - Sensor

5. Open a new window  and press the F8 key. Make the settings according to the picture on the left. Repeat the procedure with the settings as shown in the picture on the right.



Read/Write Definition

Slave ID: 1 OK

Function: 03 Read Holding Registers (4x) Cancel

Address: 0 Protocol address. E.g. 40011 -> 10

Quantity: ?

Scan Rate: 1000 [ms] Apply

Disable

☐ Read/Write Disabled

☐ Disable on error Read/Write Once

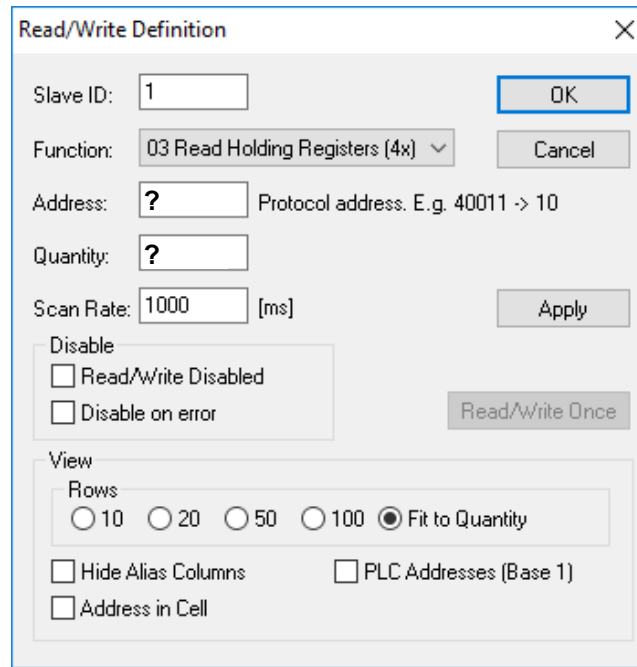
View

Rows

☐ 10 ☐ 20 ☐ 50 ☐ 100 ☒ Fit to Quantity

☐ Hide Alias Columns ☐ PLC Addresses (Base 1)

☐ Address in Cell



Read/Write Definition

Slave ID: 1 OK

Function: 03 Read Holding Registers (4x) Cancel

Address: ? Protocol address. E.g. 40011 -> 10

Quantity: ?

Scan Rate: 1000 [ms] Apply

Disable

☐ Read/Write Disabled

☐ Disable on error Read/Write Once

View

Rows

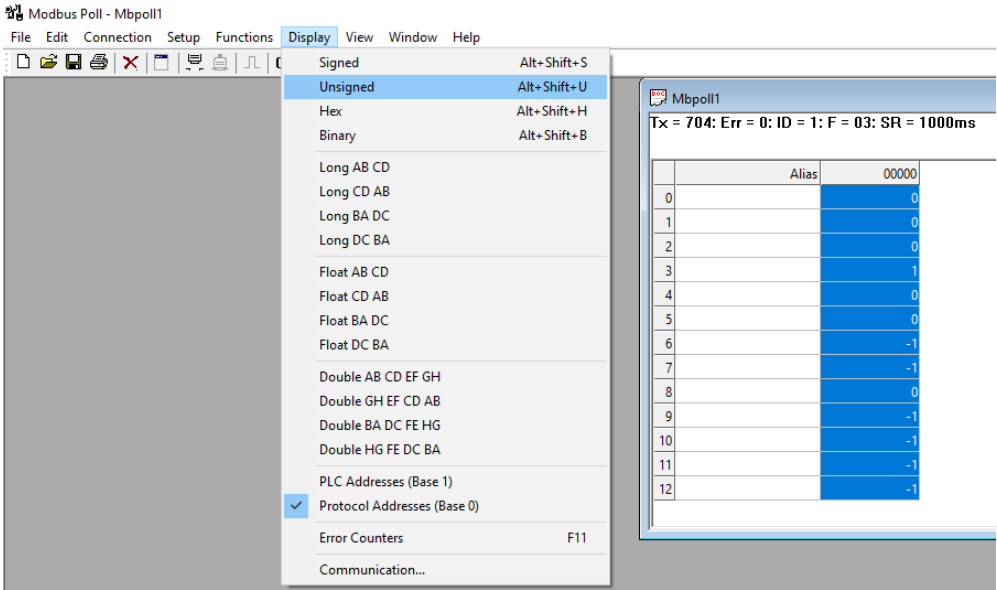
☐ 10 ☐ 20 ☐ 50 ☐ 100 ☒ Fit to Quantity

☐ Hide Alias Columns ☐ PLC Addresses (Base 1)

☐ Address in Cell

B2 - Modbus - Workshop I - Sensor

6. Display all values as *Unsigned Integer*.



7. What are the main Modbus blocs/section? e.g. in register No. 1 – 54 you find measured values

1. <i>Measured values</i>	2.	3.
4.	5.	6.
7.	8.	9.

B2 - Modbus - Workshop I - Sensor

9. Can you read out the serial number?

10. Read out the current room temperature, relative and absolute humidity

1. Value:	Interpretation:
-----------	-----------------

2. Value:	Interpretation:
-----------	-----------------

3. Value:	Interpretation:
-----------	-----------------

11. Change the unit to *Imperial*.

12. Read out the current room temperature

Value:	Interpretation:
--------	-----------------

B2 - Modbus - Workshop I - Sensor

13. Display sensor values accordingly.

CO2	rH
ppm	%
1278	63
Temp	
°C	
23.7	

14. Change the Modbus address to 60. What do you need to do:
